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# MAARULA CLASSES

# JAMIA-2026
## Official Paper

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Under the Guidence of
**Amit Katiyar**

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**AMIT KATIYAR**
(MCA-JNU)

☎ 0512-3163515 9935985550

1. Which of the following is the percentage increase in the area of a square if its side is increased by 20%?
(a) 42% (b) 43%
(c) 44% (d) 20%

2. A number is decreased by 20% and then increased by x%. What value of x will leave the number unchanged?
(a) x = 26% (b) x = 15%
(c) x= 20% (d) x = 25%

3. A dishonest dealer uses a false weight of 800 g instead of 1 kg and claims to sell at cost price. What is his profit percentage?
(a) 24% (b) 25%
(c) 26% (d) 27%

4. If the MRP of an item is 40% above the cost price and the seller gives a 20% discount on the MRP, what is the profit percentage on this item?
(a) 10% (b) 11%
(c) 12% (d) 13%

5. A seller sells two items at the same selling price. On one, he gets a profit of 40%, and on the other he incurs a loss of 20%. What is his overall result?
(a) No profit no loss
(b) 20% profit
(c) about 1.82% loss
(d) about 1.82% profit

6. The difference between the compound interest and the simple interest for 2 years at an interest rate of 10% per annum on a principal is Rs. 100. Find the principal amount. (Note: Compound interest is calculated annually.)
(a) Rs. 10,000 (b) Rs. 11,000
(c) Rs. 11,500 (d) Rs. 10,500

7. The compound interest on a principal amount for 2 years is equal to the simple interest on the same principal amount for 3 years at an interest rate of 7% per annum. Find the rate of compound interest. (Note: Compound interest is calculated annually.)
(a) 9% (b) 10%
(c) 11% (d) 12%

8. A car travels the first one-third of the distance at 20 km/h, the second one-third distance at 60 km/h, and the remaining distance at 30 km/h. What is its average speed?
(a) 25 km/h (b) 30 km/h
(c) 31 km/h (d) 32 km/h

9. A train moving at a speed of 60 km/h crosses a man walking at 6 km/h in the same direction in 15 seconds. What is the length of the train?
(a) 225 m (b) 226 m
(c) 220 m (d) 250 m

10. X, Y, and Z together can complete a work in 6 days. X alone takes 12 days and Y alone takes 18 days to complete the same work. In how many days will X and Z together complete this work?
(a) 7 (b) 8
(c) 10 (d) 9

11. If X completes half of a work in 9 days, and Y completes one-third of the same work in 12 days. In how many days will X and Y together complete this work? 10
(a) 10 (b) 11
(c) 12 (d) 13

12. How many 3-digit numbers can be formed using the digits 1, 2, 3, 4, and 5 without repetition?
(a) 60 (b) 61
(c) 62 (d) 63

13. How many different words of length 3 can be formed using all the letters of the word "BANANA"?
(a) 17 (b) 18
(c) 19 (d) 20

14. The following data are represented by a bar chart. What is the percentage increase in the height of the last bar relative to the first bar?

| Year | 2020 | 2021 | 2022 | 2023 |
|---|---|---|---|---|
| Revenue (in Lakh) | 80 | 100 | 120 | 140 |

(a) 70% (b) 73%
(c) 74% (d) 75%

15. The following data are represented by a pie chart. What is the angle of the sector (slice) corresponding to the Python programming language?

| **Programming Language** | C++ | Java | Python | Perl |
|---|---|---|---|---|
| **No of students Likes** | 80 | 100 | 120 | 140 |

(a) 107 degree (b) 108 degree
(c) 109 degree (d) 90 degree

16. If $(1011)_b = (131)_{10}$, then the base b is .........
(a) 2 (b) 3
(c) 5 (d) 7

17. If $(X)_2 = (Y)_s$, which of the following could be the value of Y if X has two 1s and one 0.
(a) 10 (b) 12
(c) 13 (d) 14

18. Which of the following is NOT a decimal number equivalent of a 5-bit binary number containing three 1s?
(a) 21 (b) 23
(c) 26 (d) 22

19. What is the value of $x$ such that the LCM (Least Common Multiple) of $x$ and 75 is 525 and their HCF (Highest Common Factor) is 5?
(a) 37 (b) 33
(c) 34 (d) 35

20. Which of the following inequalities is correct if $x$ = LCM (a, b)?
(a) max (a, b) ≤ x ≤ (a × b)
(b) a ≤ x ≤ (a × b)
(c) min(a, b) ≤ x ≤ max(a, b)
(d) a ≤ x ≤ max (a, b)

21. For what value(s) of a does the quadratic equation $ax^2 + 8x + 2 = 0$ have two real roots?
(a) a > 8 and a ≠ 0 (b) a < 8 and a ≠ 0
(c) a = 8 (d) a = 9

22. For what value of k, one root of $kx^2 - 17x + 8 = 0$ be reciprocal of the other?
(a) 17 (b) -17
(c) 8 (d) -8

23. For what value of k, the k+1, k+7, 2k is in AP?
(a) 7 (b) 6
(c) 13 (d) 14

24. Which of the following is an arithmetic progression (AP)?
(a) 2, 4, 8, 16, ...
(b) 2, 5/2, 3, 7/2, ...
(c) 0.2, 0.22, 0.222, 0.2222, ...
(d) 0.2, 0.4, 0.8, 1.0, ...

25. How many positive terms are there in the series: 75, 72, 69, ..., -6, -9, -12?
(a) 25 (b) 26
(c) 30 (d) 31

26. What is the value of sin 30° − cos 30°?
(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{\sqrt{3}-1}{2}$
(c) $\frac{\sqrt{3}+1}{2}$ (d) $\frac{1-\sqrt{3}}{2}$

27. If 3 sin θ + 4cos θ = 5, then value of sin is equal to ............
(a) 5/7 (b) 3/5
(c) 1/3 (d) 1

28. If the radius of a circle is increased by 20%, what is the percentage increase in its area?
(a) 44% (b) 45%
(c) 46% (d) 42%

29. A box has 5 blue, 2 white, and 3 red marbles. If one marble is picked at random, what is the probability that it is red?
(a) 5/10 (b) 1/10
(c) 2/10 (d) 3/10

30. Which of the following can be the probability of an event?
(a) -1/7 (b) 5/2
(c) 0.83 (d) -0.56

31. Which of the following is the equation of the line that passes through the point (2, 5) and is parallel to the line y = x?
(a) y = x + 3 (b) y = 0
(c) x = 2 (d) y = 5

32. For what value of k are the points (0, k), (-3, 0), and (5, 8) collinear
(a) 2 (b) 3
(c) 4 (d) 5

33. For what value of k, the system of equations x + 2y = 7, 5x + 10y + k = 0 has infinite number of solutions?
(a) 7 (b) -7
(c) 35 (d) -35

34. The sum of digits of a two digit number is 15. The number obtained by interchanging the digits exceeds the given number by 9. The number is.
(a) 96 (b) 87
(c) 78 (d) 69

35. The line segment joining the mid points of adjacent sides of a square form a ...........
(a) Rectangle (b) Triangle
(c) Square (d) Rhombus

36. If the base of a right-angled triangle is doubled, what should happen to its height so that the area remains unchanged?
(a) 1/2 (b) 2
(c) 1/4 (d) 1/3

37. Which of the following is not a polynomial?
(a) $x^2 + \sqrt{7}x$ (b) $\sqrt{x} + 5x$
(c) $x^2 + 3x + \sqrt{7}$ (d) $x^2 + \sqrt{3}x + \sqrt{7}$

38. Which of the following is a quadratic polynomial?
(a) $\sqrt{x^2} - 3x + 4$ (b) $\sqrt{x} + 5$
(c) $x^3 + 2x$ (d) $x^2 + \sqrt{5} - \sqrt{3}$

39. If α and β are the roots of $3x^2 + 9x = 12$, then the value of (αβ) is
(a) 3 (b) -3
(c) 4 (d) -4

40. Which of the following quadratic equation has only one root (zero)?
(a) $2x^2 + 4x + 2 = 0$ (b) $2x^2 + 4x = 2$
(c) $x^2 + 4x = 2$ (d) $2x^2 + 8x + 4 = 0$

41. Ali introduces Sidra as the sister of the only son of his father's wife. How is Sidra related to Ali?
(a) Niece (b) Mother
(c) Sister (d) Aunty

42. If AB mean 12, BC mean 23, BD mean 24, then CAB is ..............
(a) 312 (b) 313
(c) 212 (d) 213

43. Which of the following is the code of MAN in a coding scheme in which BAG is coded as 10?
(a) 13114 (b) 26
(c) 27 (d) 28

44. Which of the following is the code of BAT in a coding scheme in which CAT is coded as 16?
(a) 16 (b) 17
(c) 18 (d) 19

45. Which is the next term of the series:

**BAT, DYW, FWZ, …… ?**

(a) HUC (b) HVB
(c) HVC (d) HUB

46. Which of the following is the next number in the series

**1, 3, 6, 10, …?**

(a) 13 (b) 14
(c) 15 (d) 16

47. If f(54, 43) = 2, f(60, 51) = 10, f(70, 61) = 12, then f(72, 62) = ?

(a) 13 (b) 14
(c) 8 (d) 9

48. If $f(10, 30) = 3$, $f(60, 21) = 18$, $f(21, 11) = 6$, then $f(72, 12) = ?$

(a) 26 (b) 27
(c) 28 (d) 29

49. X is standing in the first position of a queue, and Y is four positions after X. Z is exactly in the middle of X and Y and is at the 7$^{th}$ position from the end of the queue. How many personse are there in the queue?

(a) 7 (b) 8
(c) 9 (d) 10

50. In a queue of 117 persons, if Asif's position from the beginning is 50, what is his position from the end?

(a) 66 (b) 68
(c) 67 (d) 69

51. Wax is related to Grease in the same way as Milk is related to …….?

(a) Butter (b) Drink
(c) Protein (d) Juice

52. Paw is related to Cat in the same way as Hoof is related to …….?

(a) Elephant (b) Lion
(c) Tiger (d) Horse

53. Which term of the series 2, 5, 8, 12, 14, 17, 20 is a wrong term?

(a) 1$^{st}$ (b) 2$^{nd}$
(c) 3$^{rd}$ (d) 4$^{th}$

54. Which of the following is an odd figure?

(a) A (b) E
(c) Z (d) N

55. Which of the following is correctly spelt English word?

(a) Fariegn (b) Foregn
(c) Foreign (d) Forein

56. Which of the following is correctly spelt English word?

(a) Temperature (b) Temperatur
(c) Lieutenant (d) Lifutenant

57. The commentator says that Barmy Army sings on and on. What is synonym of "Barmy"?

(a) Balmy (b) Silly
(c) Pathetic (d) Sensible

58. Employees who are diligent often get promoted faster. What is synonym of "diligent"?

(a) hardworking (b) fool
(c) cool (d) fawner

59. The garden is full of beautiful flowers. What is the part of speech (POS) of the word 'beautiful' in this sentence?

(a) Noun (b) Pronoun
(c) Adjective (d) Adverb

60. The dog barked loudly at the stranger. What is the part of speech (POS) of the word 'loudly in this sentence?

(a) Noun (b) Pronoun
(c) Adjective (d) Adverb

61. Which of the following symbols is used to represent a decision in a flowchart?

(a) Rectangle (b) Diamond
(c) Parallelogram (d) Oval

62. Which of the following sorting algorithms is based on the insertion or bubble sort algorithm?
(a) Quick sort algorithm
(b) Merge sort algorithm
(c) Radix sort algorithm
(d) Shell sort algorithm

63. Which of the following is an interpreted programming language?
(a) C (b) C++
(c) Python (d) Pascal

64. Which of the following programming languages allows overloading of the * operator in a user-defined class 'Vector' to support scalar multiplication?
(a) C (b) C++
(c) Java (d) Pascal

65. Which of the following is a Database Management System (DBMS)?
(a) Ingres (b) Bing
(c) Baidu (d) Safari

66. Which of the following types of memory is a part of the CPU?
(a) RAM (b) ROM
(c) Cache (d) Register

67. What is the main purpose of virtual memory?
(a) To increase CPU speed
(b) To replace RAM by Cache
(c) To increase RAM size logically
(d) To reduce the uses of registers

68. Which of the following is an invalid cell reference in MS Excel?
(a) 1C (b) C$1
(c) $C1 (d) $C$1

69. Which of the following MS Excel formulas displays the day name in short form (e.g., Mon, Tue, Thu) for the current day?
(a) = TEXT(TODAY(),"dddd")
(b) = TEXT(TODAY(),"ddd")
(c) = DAY(TODAY(),"dddd")
(d) = DAY(TODAY(),"ddd")

70. Which of the following should be an invalid IP address in IPv4?
(a) 169.254.0.1 (b) 172.31.255.253
(c) 169.254.254.254 (d) 172.260.255.253

71. Which of the following is not a peripheral device?
(a) Joystick (b) Touch-Screen
(c) ALU (d) Mouse

72. Which of the following is an input device?
(a) Projector (b) Motherboard
(c) Printer (d) Scanner

73. Which of the following is a non-preemptive multitasking operating system?
(a) Windows NT (b) Windows 3.1
(c) Unix (d) Linux

74. In which generation of computers is the pipelining technique used?
(a) First generation
(b) Second generation
(c) Third generation
(d) Fourth generation

75. Which of the following is not a web-browser?
(a) Google (b) Google Chrome
(c) Microsoft Edge (d) Opera

76. Which of the following is a web-server?
(a) Opera (b) NGINX
(c) Yandex (d) Baidu

77. Which of the following is an example of the firmware?
(a) BIOS
(b) Compilers
(c) Operating Systems
(d) Interpreters

78. Which algorithm design technique can be used to find all longest common subsequences (LCSs) of two given sequences?
(a) Greedy method
(b) Divide and Conquer
(c) Backtracking
(d) Dynamic Programming

79. What is the best-case time complexity of the binary search algorithm?
(a) O(1) (b) O(n)
(c) O(log n) (d) O(nlog n)

80. Which of the following is **NOT** a characteristic of an algorithm?
(a) Input (b) Output
(c) Scalable (d) Definiteness

81. If ${}^{7}C_x = {}^{7}C_{x-1}$, then $x$ is:
(a) 7 (b) 2
(c) 3 (d) 4

82. For what value of n, $\left(x+\frac{1}{x}\right)^7$ and $(1+x)^n$ have same coefficient of x? 7
(a) 5 (b) 34
(c) 35 (d) 36

83. Which of the following is the sum of the series

$$\mathbf{1 + 2 + 3 + \ldots + \frac{(n-2)}{2}?}$$

(a) $\frac{n\times(n-2)}{8}$ (b) $\frac{n\times(n-2)}{4}$
(c) $\frac{n\times(n-2)}{2}$ (d) $\frac{n\times(n-1)}{8}$

84. The value of the $\sum_{i=0}^{n-1}\left[-\frac{i^2}{2}-\frac{i}{2}+\frac{n(n+1)}{2}\right]$ is ........
(a) $\frac{n(n+1)(2n+1)}{6}$ (b) $\frac{n(n+1)(n+2)}{6}$
(c) $\frac{n(n+1)(n-2)}{6}$ (d) $\frac{n(n+1)(n+2)}{4}$

85. If A = $\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$, then $A^{37} = \cdots$
(a) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ (b) $\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$
(c) $\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$ (d) $\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix}$

86. How many reflexive relations are possible on the set A = {1, 2, 3, 4)?
(a) 4 (b) 2048
(c) 1024 (d) 4096

87. Which of the following is an odd function?
(a) $f(x) = 1 - x^2$
(b) $f(x) = x - cos^3 x$
(c) $f(x) = x - \sin x$
(d) $f(x) = x^3 - |x|$

88. Which of the following function is defined on set of integers?
(a) $f(x) = \sqrt{7-x}$ (b) $f(x) = \sqrt{x^2}$
(c) $f(x) = \sqrt{7-x^2}$ (d) $f(x) = \sqrt{x^3}$

89. Let z = x + iy be a complex number, then the locus represented by |z + 1| = |z + i| is a:
(a) Circle: $x^2 + y^2 = 1$
(b) Ellipse: $\frac{x^2}{2^2} + \frac{y^2}{i^2} = 1$
(c) Straight line: x + y = 0
(d) Straight line: x - y = 0

90. What is the value of $\left(i^{13} - \left(\frac{1}{i}\right)^{13}\right)^2$ ?
(a) 4 (b) -4
(c) 4i (d) -4i

91. If A and B are two matrices such that AB = B and BA = A, then $A^2 + B^2$ = .........
(a) AB (b) A+B
(c) A (d) 2AB

92. If A = $\begin{bmatrix}0 & x\\ 2 & 0\end{bmatrix}$ and $|A^2| = 100$, then value of x is:
(a) $\pm 2$ (b) $\pm 3$
(c) $\pm 4$ (d) $\pm 5$

93. If $\begin{vmatrix}1! & 2! & 3!\\ 2! & 3! & 4!\\ 3! & 4! & 5!\end{vmatrix} = x$, then value of x is ......
(a) 2! (b) 3!
(c) 4! (d) 5!

94. $\lim_{x\to\infty} \frac{\sqrt{x}}{\sqrt{x+\sqrt{x+\sqrt{x}}}}$ equals to
(a) 0 (b) 1
(c) 2 (d) ∞

95. $\lim_{x\to\infty} \frac{5x^2+7}{2x^2+3x}$ is equals to:
(a) 5/2 (b) 7/3
(c) 5 (d) 5/3

96. If y = $cos^{-1}x$ and z = $sin^{-1}\sqrt{1-x^2}$, then $\frac{dy}{dz}$ is equal to ..................
(a) $1/(1-x^2)$ (b) $x/(1+x^2)$
(c) 1/2 (d) 1

97. If $\sqrt{x+y}+\sqrt{x-y}=\sqrt{2}$, then $\frac{dy}{dx}$ is equal to ...

(a) $\sqrt{x+y}+\sqrt{x-y}$ (b) $\frac{1}{\sqrt{x+y}}+\frac{1}{\sqrt{x-y}}$

(c) $1/y$

(d) $1/x$

98. If $\int_0^{\pi/3}\frac{\cos x}{3+4\sin x}dx=\frac{1}{4}log k$, then k = .........

(a) $\left(1+\frac{2}{\sqrt{3}}\right)$

(b) $\left(3+\sqrt{3}\right)$

(c) $\left(3+\sqrt{2}\right)$

(d) $\left(2+\sqrt{3}\right)$

99. $\int\frac{dx}{x\log x\log(\log x)}$ is equal to............

(a) $\log x$ (b) $\log(\log x)$

(c) $\left(\log(\log x)\right)^2$ (d) $\log\left(\log(\log x)\right)$

100. Lot $\vec{A}=i-j+k, \vec{C}=-i-j$ be two vectors. Which of the following is the vector $\vec{B}$ such that $\vec{A}\times\vec{B}=\vec{C}$ and $\vec{A}.\vec{B}=1$?

(a) $i+j$

(b) $k$

(c) $i+k$

(d) $j+k$

## ANSWER KEY

| | | | | |
|---|---|---|---|---|
| 1. | 2. | 3. | 4. | 5. |
| c | d | b | c | d |
| 6. | 7. | 8. | 9. | 10. |
| a | b | b | a | d |
| 11. | 12. | 13. | 14. | 15. |
| c | a | c | d | b |
| 16. | 17. | 18. | 19. | 20. |
| c | a | b | d | a |
| 21. | 22. | 23. | 24. | 25. |
| b | c | c | b | a |
| 26. | 27. | 28. | 29. | 30. |
| d | b | a | d | c |
| 31. | 32. | 33. | 34. | 35. |
| a | b | d | c | c |
| 36. | 37. | 38. | 39. | 40. |
| a | b | d | d | a |
| 41. | 42. | 43. | 44. | 45. |
| c | a | b | d | c |
| 46. | 47. | 48. | 49. | 50. |
| c | d | b | c | b |

| | | | | |
|---|---|---|---|---|
| 51. | 52. | 53. | 54. | 55. |
| a | d | d | b | c |
| 56. | 57. | 58. | 59. | 60. |
| a | b | a | c | d |
| 61. | 62. | 63. | 64. | 65. |
| b | a | c | c | a |
| 66. | 67. | 68. | 69. | 70. |
| d | c | a | b | d |
| 71. | 72. | 73. | 74. | 75. |
| c | d | b | d | a |
| 76. | 77. | 78. | 79. | 80. |
| b | a | d | a | c |
| 81. | 82. | 83. | 84. | 85. |
| d | c | a | a | c |
| 86. | 87. | 88. | 89. | 90. |
| d | c | c | d | b |
| 91. | 92. | 93. | 94. | 95. |
| b | – | c | b | a |
| 96. | 97. | 98. | 99. | 100. |
| d | c | a | d | b |